n=41 cold end-to-end timing run

Session `ECBS1hb77d7527b579` measured the current B2 and B3 six-sum pipelines from cold setup through verified scalar recovery against the matched strong-rho reference. All 24 planned executions verified: four warmups and 20 measured runs, five per arm. Exact replay reproduced all 20 measured deterministic runs.

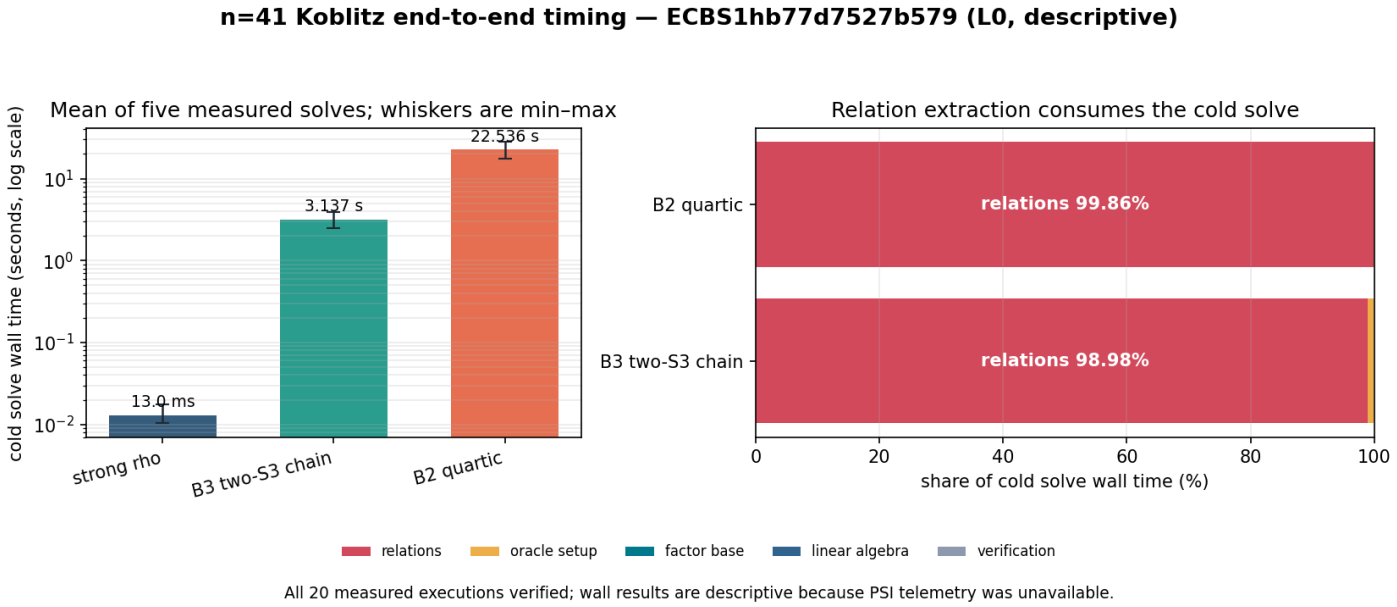


Figure 1: Cold solve time and phase attribution

Run identity

field	recorded value
UTC start	2026-10-13 22:07:16
spec	`ECS1hc7fc552628ee`, `E2E_SPEC.json`
workload	`W2d6c43bf8940`, public target on `icv1-f2m41-tm2308219-7f48b14a`
source revision	`5f83882f976499a55a1f6851281131a85f7611db`, clean at build
binary SHA-256	`30050604...44d2a1` (full value below)
host	Intel Xeon Platinum 8573C, virtualized Linux, CPU 4 reserved
isolation	L0 for all measured runs
audit	`audit-replay-all.json`; 20/20 measured replays identical, zero problems

Binary SHA-256: `3005060405a6e2a0cab404b2023b07d7c023d44f2a5c47feeb7f157f5944d2a1`. Audit receipt SHA-256: `46f230bbd9d91d8b79745a3701d547ace0f434bf2daa4232291c4d95034ffcfd`.

The strict command without `--allow-busy` refused before executing because CPU and memory PSI telemetry were unavailable. The unchanged spec was then run with the explicit downgrade:

```
./target/release/ecbench run --wait --allow-busy --cpus auto \
  --spec research/koblitz_higher_arity_rho_20261013/E2E_SPEC.json \
  --out research/koblitz_higher_arity_rho_20261013/sessions/n41-e2e-wall-l0-1
```

The run also recorded unmovable user threads, hypervisor steal in some runs, and no PMU. Consequently all elapsed-time values below are descriptive rather than admitted wall claims.

Recorded elapsed time

`cold solve` begins with fresh-process method setup and ends after internal scalar verification. `process` adds fork/exec, teardown and record transport. `online` is the separately recorded target-dependent window.

arm	measured n	cold solve mean	cold solve median [min, max]	process mean	online mean
strong rho	5	13.005 ms	11.904 [10.354, 17.549] ms	34.053 ms	10.320 ms
B3 two-S3 chain	5	3.137 s	3.126 [2.461, 3.933] s	3.185 s	3.106 s
B2 quartic	5	22.536 s	22.283 [17.604, 27.906] s	22.588 s	22.505 s

Derived from the recorded sums, B3 consumed `0.1392x` B2's cold solve time (`7.183x` less wall time). B3 consumed `241.237x` rho's cold solve time; the fork-to-reap ratio was `93.535x` because fixed process overhead is large beside a 13 ms rho solve. Ecbench's paired descriptive median ratios were `0.1398` for B3/B2 and `279.895` for B3/rho. These are observations on one L0 host and one public target, not admitted wall-speed claims.

Exclusive IC phase clocks

phase	B2 mean	B2 share of solve	B3 mean	B3 share of solve
factor base	0.616 ms	0.003%	0.594 ms	0.019%
oracle setup	27.691 ms	0.123%	28.523 ms	0.909%
relations	22.504 s	99.858%	3.105 s	98.981%
linear algebra	0.752 ms	0.003%	0.366 ms	0.012%
verification	0.015 ms	<0.001%	0.015 ms	<0.001%

The phase clock recorded `7.247x` less relation wall time for B3. The other phases are effectively unchanged at this scale.

Relation-work counters

recorded counter across five measured runs	B2	B3
trials / verified relations	98 / 98	98 / 98
pair cells visited	10,128,845	10,128,845
mean pair cells per solve	2,025,769	2,025,769
pair cells per verified relation	103,355.6	103,355.6
table lookups	40,515,237	40,515,237
root solves	10,128,845 S4	20,257,690 S3
relation wall per pair cell	11.109 us	1.533 us

The counters and relation/rank results match between B2 and B3. The replacement changes the cost of processing a cell; it does not change how many cells the pipeline scans.

Evidence and limits

- Raw sealed session: `sessions/n41-e2e-wall-l0-1`
- Resource aggregate: `resources.json`
- Replay receipt: `audit-replay-all.json`
- Strict L2 refusal: `e2e-l2-refusal.stderr.txt`
- Frozen protocol: `E2E_PROTOCOL.md`
- Graph data and source: `data.json`, `plot.py`

This is one n=41 workload and five rounds on one host. It is not a scaling experiment, an independent-host timing, or evidence for ECC2K-130. B2's S4 solves, B3's S3 solves, lookups, and several factor-base operations remain unpriced in `S`; counted-work ratios are lower-bound diagnostics and are not substitutes for the elapsed-time observations above.